100 Expected Q&A Questions

Case Study 5 — Multiple Regression in Biology & Analytics

STAT 482 | Mohammed bin Salem | Spring 2026

How to use this: Read the question, cover the answer, try to say it in your own words. If you get it wrong, mark it and review again. The answers are short on purpose — say exactly that.

Categories: 1–15 Dataset & Variables | 16–28 EDA & Cleaning | 29–45 Model Building | 46–57 Model Selection | 58–72 Diagnostics | 73–82 Validation | 83–93 Interpretation | 94–100 R & Packages

📋 Dataset & Variables

#	Question	Short Answer
1	**What dataset did you use?**	The Forest Fires dataset from the UCI Machine Learning Repository.
2	**How many observations does the dataset have?**	517 observations (fire events).
3	**How many variables are in the dataset?**	13 total — 8 predictors we used, plus spatial, temporal, and response variables.
4	**What is the response variable?**	Burned area (area) — measured in hectares.
5	**Why did you use burned area as a proxy for ecosystem stress?**	Because a larger burned area means more environmental damage to the ecosystem.
6	**What are the 8 predictor variables?**	FFMC, DMC, DC, ISI, temperature, relative humidity, wind speed, and rainfall.
7	**What does FFMC stand for?**	Fine Fuel Moisture Code — it measures moisture in fine surface fuels like dry leaves.
8	**What does DMC stand for?**	Duff Moisture Code — it measures moisture in loosely packed organic soil layers.
9	**What does DC stand for?**	Drought Code — it measures long-term drought and deep soil dryness.
10	**What does ISI stand for?**	Initial Spread Index — it estimates how fast a fire can spread.
11	**What system do FFMC, DMC, DC, and ISI belong to?**	The Canadian Fire Weather Index (FWI) system.
12	**Were there any missing values in the dataset?**	No — zero missing values. The dataset was complete.
13	**Where is the data from geographically?**	Montesinho Natural Park in northeast Portugal.
14	**What does the burned area variable look like?**	It is heavily right-skewed — most fires burned 0 ha, but a few burned over 1,000 ha.
15	**What is the mean burned area?**	12.85 hectares.

🔍 Exploratory Data Analysis & Cleaning

#	Question	Short Answer
16	**Did you need to clean the data?**	Not much — no missing values. We converted month and day to ordered factors.
17	**What transformation did you apply to the response variable?**	We used log(area + 1) to reduce the skewness.
18	**Why did you add 1 before taking the log?**	Because log(0) is undefined, and many fires had 0 ha burned.
19	**What is a right-skewed distribution?**	Most values are small, but a few very large values pull the tail to the right.
20	**Why is skewness a problem for regression?**	Because OLS regression assumes the errors are approximately normally distributed.
21	**What did the boxplots show?**	Many extreme outliers — especially in the area variable.
22	**What was the highest correlation between any predictor and log_area?**	0.07 — which is very weak.
23	**What does a correlation of 0.07 mean?**	Almost no linear relationship between that predictor and the burned area.
24	**Which two predictors were most correlated with each other?**	DMC and DC — with a correlation of 0.68.
25	**What was the correlation between temperature and relative humidity?**	−0.53 — negative, meaning when temperature goes up, humidity goes down.
26	**What did the scatterplots show?**	Almost flat trend lines — no predictor has a strong individual effect on burned area.
27	**What does EDA stand for?**	Exploratory Data Analysis.
28	**Why do we do EDA before building the model?**	To understand the data, find problems, and decide if transformations are needed.

📐 Model Building

#	Question	Short Answer
29	**What type of model did you build?**	Multiple Linear Regression (OLS).
30	**What does OLS stand for?**	Ordinary Least Squares — it minimizes the sum of squared errors.
31	**How many models did you compare?**	Four: Full model, Time Controls model, Interaction model, and Stepwise model.
32	**What is the full model?**	A regression with all 8 predictors included.
33	**What was the R-squared of the full model?**	0.020 — or about 2%.

#	Question	Short Answer
34	**What was the Adjusted R-squared of the full model?**	0.004 — less than 1%.
35	**What does R-squared mean?**	It tells us what percentage of variation in the response is explained by the model.
36	**What does Adjusted R-squared mean?**	Same as R-squared but it penalizes for adding extra predictors that don't help.
37	**Why is Adjusted R-squared better than R-squared?**	Because R-squared always increases when you add more variables, even useless ones.
38	**What was the F-statistic p-value for the full model?**	0.247 — not significant.
39	**What does a non-significant F-test mean?**	The model as a whole does not explain the response significantly.
40	**Which variable was the only significant predictor?**	Wind speed — with a p-value of 0.039.
41	**What was the wind coefficient in the full model?**	0.076 — positive, meaning more wind = larger fire.
42	**Why does wind increase burned area?**	Wind dries out fuel, supplies oxygen, and carries embers to spread fire faster.
43	**What is the interaction model?**	A model that adds a temp × RH interaction term to test if their combined effect matters.
44	**Was the interaction term significant?**	No — p-value was 0.446.
45	**What is the time controls model?**	A model that adds month and day as categorical variables to control for seasonality.

🏆 Model Selection & Stepwise

#	Question	Short Answer
46	**What is stepwise regression?**	An automatic method that adds or removes variables to find the best combination.
47	**What direction was the stepwise used?**	Both directions — it can add and remove variables.
48	**Which R function did you use for stepwise?**	stepAIC() from the MASS package.
49	**Which three variables did stepwise keep?**	DMC, RH (relative humidity), and wind speed.
50	**What was the AIC of the stepwise model?**	1815.52 — the lowest of all models.
51	**What does AIC stand for?**	Akaike Information Criterion.
52	**What does a lower AIC mean?**	A better balance between model fit and simplicity.
53	**What is BIC?**	Bayesian Information Criterion — similar to AIC but penalizes complexity more.
54	**What was the BIC of the stepwise model?**	1836.76 — also the lowest.

#	Question	Short Answer
55	**Why is the stepwise model considered the best?**	It has the lowest AIC and BIC — best fit with fewest variables.
56	**What was the Adjusted R-squared of the time controls model?**	0.021 — the highest, but its AIC and BIC were worse due to too many variables.
57	**What is the stepwise model equation?**	log(area+1) = 0.913 + 0.00175·DMC − 0.00558·RH + 0.0624·wind

🔬 Model Diagnostics & Assumptions

#	Question	Short Answer
58	**What is VIF?**	Variance Inflation Factor — it measures multicollinearity between predictors.
59	**What VIF value is considered a problem?**	VIF above 10 is serious; above 5 is a concern.
60	**What were the VIF values in your model?**	All below 5 — the highest was temperature at 2.66.
61	**Is multicollinearity a problem in your model?**	No — all VIF values are acceptable.
62	**What is multicollinearity?**	When two or more predictor variables are highly correlated with each other.
63	**What is the Breusch-Pagan test used for?**	To test if the variance of residuals is constant (homoscedasticity).
64	**What was the Breusch-Pagan result?**	BP = 11.27, p = 0.187 — we fail to reject homoscedasticity. It passed.
65	**What is heteroscedasticity?**	When the variance of residuals changes across fitted values — it violates OLS assumptions.
66	**What is the Shapiro-Wilk test used for?**	To test if the residuals follow a normal distribution.
67	**What was the Shapiro-Wilk result?**	W = 0.854, p < 2.2e-16 — residuals are NOT normal.
68	**Is non-normal residuals a big problem here?**	Not critically — with N=517, the Central Limit Theorem protects our inference.
69	**What did the Q-Q plot show?**	Points deviated from the line in the upper tail — confirming non-normality.
70	**What did the Residuals vs Fitted plot show?**	A slight fan shape — some potential heteroscedasticity visually.
71	**What are Cook's distance and leverage?**	Cook's distance measures influence; leverage measures how extreme a data point is.
72	**Did any observation have dangerous influence?**	No — observation 500 had high leverage but was within Cook's distance limits.

✅ Model Validation

#	Question	Short Answer
73	**How did you validate the model?**	By splitting data into 80% training and 20% test sets.
74	**How many observations in training set?**	416 observations.
75	**How many observations in the test set?**	101 observations.
76	**Which R package did you use for the split?**	The caret package.
77	**What is RMSE?**	Root Mean Squared Error — the average size of prediction errors, squared then square-rooted.
78	**What was the RMSE on the test set?**	1.71 — on the log(area+1) scale.
79	**What is MAE?**	Mean Absolute Error — the average absolute difference between predicted and actual values.
80	**What was the MAE on the test set?**	1.30 — on the log(area+1) scale.
81	**Is an RMSE of 1.71 good or bad?**	It's not great — it roughly means about ±5.5 ha error on the original scale.
82	**Why is prediction error so high?**	Because burned area depends on many factors not in our dataset — like terrain and suppression.

💡 Interpretation & Implications

#	Question	Short Answer
83	**What does the positive wind coefficient mean biologically?**	Higher wind speed is associated with larger burned areas — wind spreads fire.
84	**What does the negative RH coefficient mean?**	Lower humidity leads to more burning — dry air dries out fuel and increases fire risk.
85	**What does DMC tell us about fire risk?**	Higher DMC means drier organic soil — which sustains longer, larger fires.
86	**Which month showed the highest fire risk?**	July — the month^7 term in the time controls model was close to significant.
87	**What is a practical implication for fire management?**	Set up wind-based alert systems — warn when wind exceeds 5–6 km/h during drought.
88	**Why can't we predict fire size perfectly from weather?**	Because terrain, vegetation type, and firefighting response also determine fire size.
89	**What additional data would improve the model?**	GIS data — slope, aspect, vegetation index, distance to roads.
90	**What does the intercept of 0.913 in the stepwise model mean?**	When DMC, RH, and wind are all zero, the predicted log_area is 0.913.
91	**Should we back-transform predictions?**	Yes — use e^(predicted) – 1 to convert back to hectares.
92	**What is the business implication of this analysis?**	Focus fire prevention resources on windy and dry days, especially in July and August.
93	**Why is the overall model still useful even with low R-squared?**	Because it identifies wind as a key driver and provides a statistically sound starting point.

💻 R Packages & Functions

#	Question	Short Answer
94	**Which R packages did you use?**	stats, car, MASS, caret, ggplot2, lmtest, GGally, dplyr, tidyverse.
95	**What is the car package used for?**	For VIF calculation — multicollinearity diagnostics.
96	**What is the MASS package used for?**	For stepwise regression using the stepAIC() function.
97	**What is the caret package used for?**	For splitting data into training and test sets.
98	**What is the lmtest package used for?**	For the Breusch-Pagan homoscedasticity test (bptest function).
99	**What is ggplot2 used for?**	For creating visualizations — scatterplots, histograms, and other charts.
100	**What does the lm() function do in R?**	It fits a linear regression model.

> *Tip: You do not need to memorize the exact numbers. But know the direction (e.g. "low R-squared", "only wind was significant", "all VIF below 5"). That is usually enough to show you understood the analysis.*